



W² Education Standards Authority

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Centre Number

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Student Number

2077

HIGHEST SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks:
70

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–12)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Given $\tan \theta = \frac{1}{2}$, what is the exact value of $\tan \left(\theta - \frac{\pi}{6} \right)$?

A. $8 + 5\sqrt{3}$

B. $8 - 5\sqrt{3}$

C. $\frac{8 + 5\sqrt{3}}{11}$

D. $\frac{8 - 5\sqrt{3}}{11}$

2 The function f is defined by the following parametric equations:

$$x = 2e^t, \quad y = \cos(1 + e^{3t}).$$

Which of the following is equivalent to $\frac{dy}{dx}$?

A. $-\frac{3}{2}e^{2t} \sin(1 + e^{3t})$

B. $\frac{3}{2}e^{2t} \sin(1 + e^{3t})$

C. $6e^{2t} \sin(1 + e^{3t})$

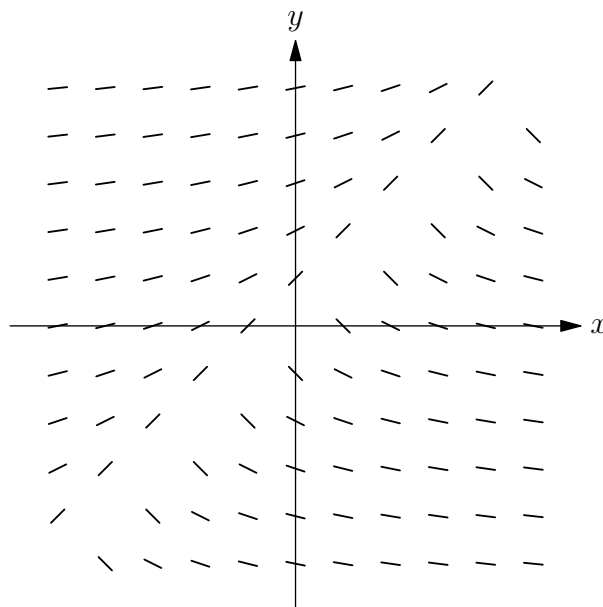
D. $-6e^{2t} \sin(1 + e^{3t})$

- 3 Daniel has recently been into collecting Labubus and wants to display a total of five different Labubus on his shelf. He can choose between five different Labubus from the *Have a Seat* collection and five different Labubus from the *Exiting Macaron* collection.

Given that the pink *Exiting Macaron* Labubu must be displayed, how many ways can he select the remaining if he wants at least two from the *Have a Seat* collection and at least one other from the *Exiting Macaron* collection on the shelf?

- A. 100
- B. 240
- C. 252
- D. 320

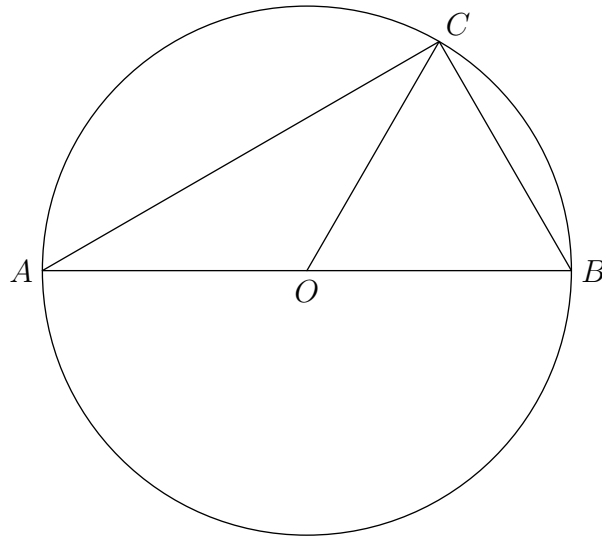
- 4 Which of the following differential equations matches the slope field shown?



- A. $\frac{dy}{dx} = y - x$
- B. $\frac{dy}{dx} = x - y$
- C. $\frac{dy}{dx} = \frac{1}{y - x}$
- D. $\frac{dy}{dx} = \frac{1}{x - y}$

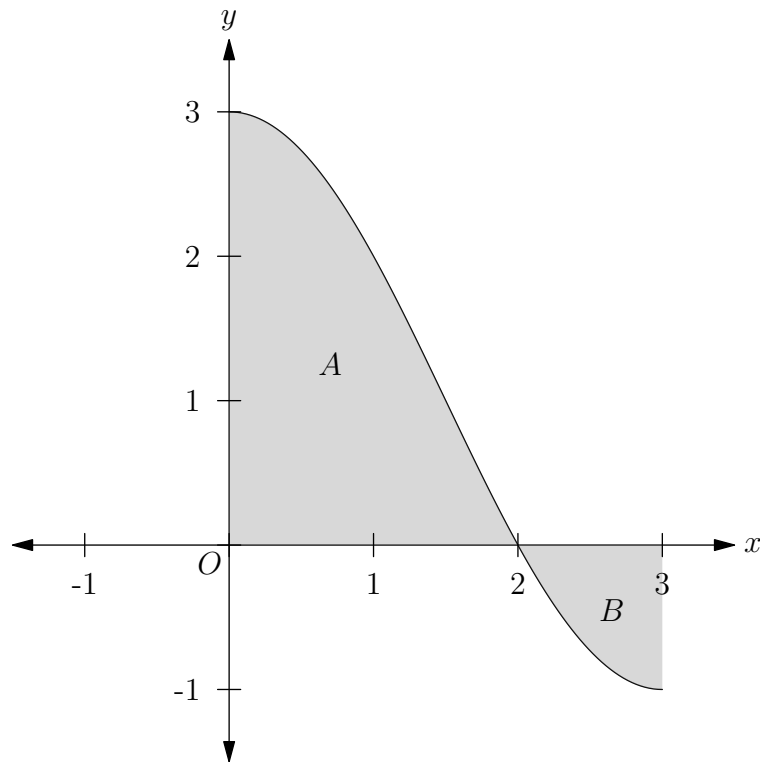
- 5 The polynomial $P(x) = 6x^4 - x^3 - 17x^2 + 16x - 4$ has four real roots. What is the sum of the roots of the polynomial $P\left(-\frac{1}{2}x - 1\right)$?
- A. $\frac{5}{6}$
- B. $\frac{1}{12}$
- C. $-2\frac{1}{3}$
- D. $-8\frac{1}{3}$
- 6 What is the constant term in the binomial expansion of $\left(4x^2 - \frac{1}{x}\right)^9$?
- A. -344064
- B. -5376
- C. 84
- D. 5376
- 7 Which of the following expressions is equivalent to $\int \frac{4}{x}(\ln x)^3 \, dx$?
- A. $(\ln x)^4 + C$
- B. $4x^4 \ln x + C$
- C. $\frac{4x^4}{\ln x} + C$
- D. $\frac{4}{x^4} + C$

- 8 In the diagram below, AOB is a diameter of the circle with centre O . Point C lies on the circumference of the circle.



- If $\overrightarrow{OC} = \mathbf{r}$ and $\overrightarrow{BC} = \mathbf{s}$, to which of the following is $\overrightarrow{AC}$ equal?
- A. $\mathbf{r} + 2\mathbf{s}$
 - B. $\mathbf{r} - 2\mathbf{s}$
 - C. $2\mathbf{r} + \mathbf{s}$
 - D. $2\mathbf{r} - \mathbf{s}$
- 9 A particle is projected from a window 9 metres above the ground, at an angle of θ above the horizontal, where $\tan \theta = \frac{3}{4}$ and the initial velocity is 20ms^{-1} . Taking $g = 10$, what is the maximum height reached by the particle?
- A. 6.5 metres
 - B. 7.2 metres
 - C. 15.5 metres
 - D. 16.2 metres

- 10 The graph of $y = 2 \cos\left(\frac{\pi x}{3}\right) + 1$ is shown.



Region A has an area of $\frac{3\sqrt{3}}{\pi} + 2$ square units.

Region B has an area of $\frac{3\sqrt{3}}{\pi} - 1$ square units.

Using this information, what is the value of $\int_{-1}^3 \frac{3}{\pi} \cos^{-1}\left(\frac{x-1}{2}\right) dx$?

- A. 3
- B. 6
- C. $\frac{6\sqrt{3}}{\pi} + 1$
- D. $11 - \frac{6\sqrt{3}}{\pi}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (12 marks) Use the Question 11 Writing Booklet

- (a) Consider the vectors $\mathbf{a} = m\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{b} = \hat{\mathbf{i}} + m\hat{\mathbf{j}}$, where $m \in \mathbb{R}$. Find the values of m for which the acute angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{3}$. 2

- (b) The letters from the word VOLUME are placed at random on the circumference of a circle without replacement.

- (i) Find the total number of distinct arrangements. 1

If one of the arrangements are chosen at random, find the probability that:

- (ii) all the vowels are together. 2

- (iii) the vowels and consonants alternate. 1

- (c) Given $P(x) = 3x^3 - 2x^2 + x - 3$ has zeros α , β and γ :

- (i) write the value of $\alpha\beta\gamma$. 1

- (ii) Hence, or otherwise, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 1

- (d) Prove by mathematical induction that, for all positive integers n , 4

$$\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \cdots + \frac{2}{n(n+2)} = \frac{3}{2} - \frac{2n+3}{(n+1)(n+2)}.$$

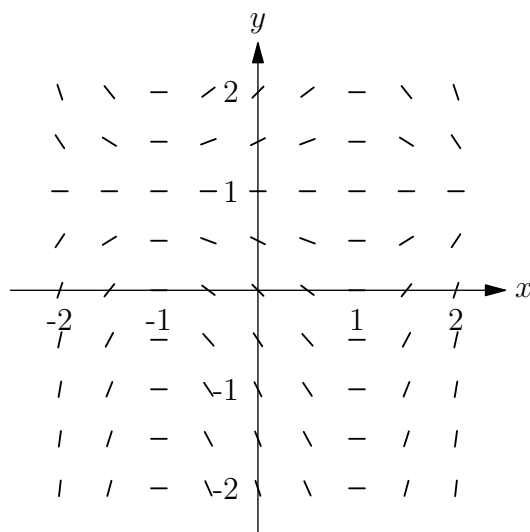
End of Question 11

Question 12 (16 marks) Use the Question 12 Writing Booklet

(a) Find $\int \sin x \sin 3x \, dx$.

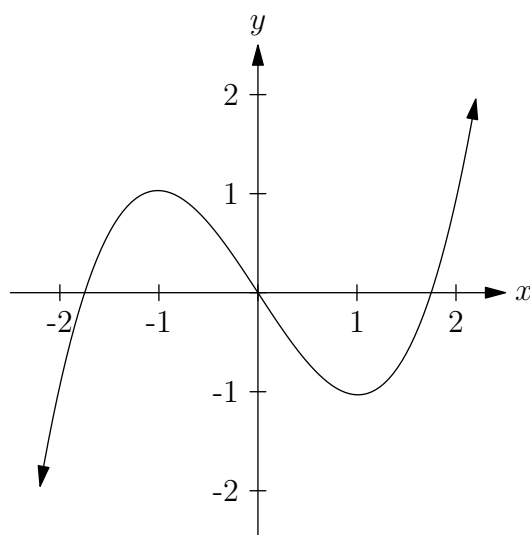
2

- (b) Consider the differential equation $\frac{dy}{dx} = -(x^2 - 1)(y - 1)$ and its slope field, which is graphed below.



- (i) By referring to the slope field for the differential equation, explain why a solution could not have the graph shown below.

1



- (ii) Find the particular solution of the differential equation that passes through the point $(0, 2)$.

3

Question 12 continues on page 9

Question 12 (continued)

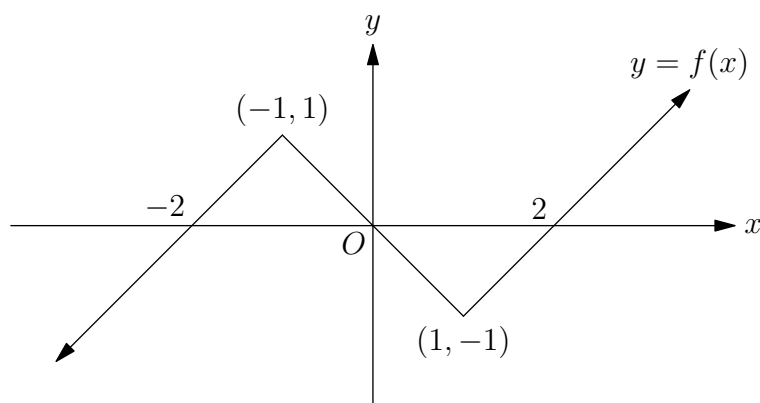
- (c) (i) Express $\sin x - \sqrt{3} \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2

- (ii) Hence, solve $\sin x - \sqrt{3} \cos x = 2$ for $0 \leq x \leq \pi$. 1

- (d) A biased coin has probability 0.7 of landing on heads when tossed. 3

What is the most likely combination of heads and tails to occur in a sequence of five tosses? Justify your answer with relevant calculations.

- (e) Consider the graph of $y = f(x)$ below.



On separate number planes, graph each of the following:

- (i) $y = \frac{1}{f(x)}$. 2

- (ii) $y = f(|x| - 2)$. 2

End of Question 12

Question 13 (14 marks) Use the Question 13 Writing Booklet

- (a) The roots of the polynomial $P(x) = x^3 + qx^2 + rx + t$ are in an arithmetic progression. Show that **3**

$$2q^3 - 9qr + 27t = 0.$$

- (b) In a square with sides of unit length, 51 points are chosen. **3**

Show that there are at least 3 points which lie within a circle of radius $\frac{1}{5\sqrt{2}}$.

- (c) Two chemicals $\mathcal{C}_1$ and $\mathcal{C}_2$ are put together in a solution, where they react to form a solid, $\mathcal{S}$. The rate of increase of the mass, m kg, of $\mathcal{S}$ is proportional to the product of the masses of unreacted $\mathcal{C}_1$ and $\mathcal{C}_2$ present at time t minutes.

It takes 1kg of $\mathcal{C}_1$ and 3kg of $\mathcal{C}_2$ to form 4kg of $\mathcal{S}$.

Initially, 2kg of $\mathcal{C}_1$ and 3kg of $\mathcal{C}_2$ are put together in solution, and 1kg of $\mathcal{S}$ forms in 1 minute.

- (i) Carefully prove that **3**

$$\frac{dm}{dt} = k(8 - m)(4 - m),$$

where k is a constant.

- (ii) Given that **3**

$$\frac{1}{(8 - x)(4 - x)} = -\frac{1}{4(8 - x)} + \frac{1}{4(4 - x)},$$

show that

$$t = -\frac{1}{\ln \frac{7}{6}} \ln \left(\frac{8 - m}{8 - 2m} \right),$$

for $0 \leq m \leq 4$.

- (iii) Hence, find the exact mass of $\mathcal{S}$ formed after 2 minutes. **2**

End of Question 13

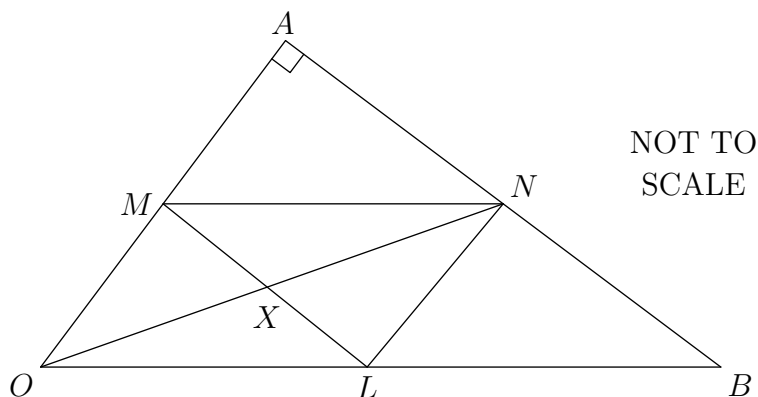
Question 14 (18 marks) Use the Question 14 Writing Booklet

- (a) The function $f(x) = \frac{1 + kx}{x^2 - 1}$ is continuous on the domain $-1 < x < 1$. **3**

Find the values of k for which the inverse function $f^{-1}(x)$ exists.

- (b) Prove that $\cos^{-1}(x) + \cos^{-1}(-x) = \pi$. **2**

- (c) In the diagram below, $\triangle AOB$ is right-angled with $\angle AOB = 90^\circ$.
The midpoints of OA , AB and OB are M , N and L , respectively.
The intersection of ML and ON is X .



Let $\underline{a} = \overrightarrow{OA}$ and $\underline{b} = \overrightarrow{OB}$.

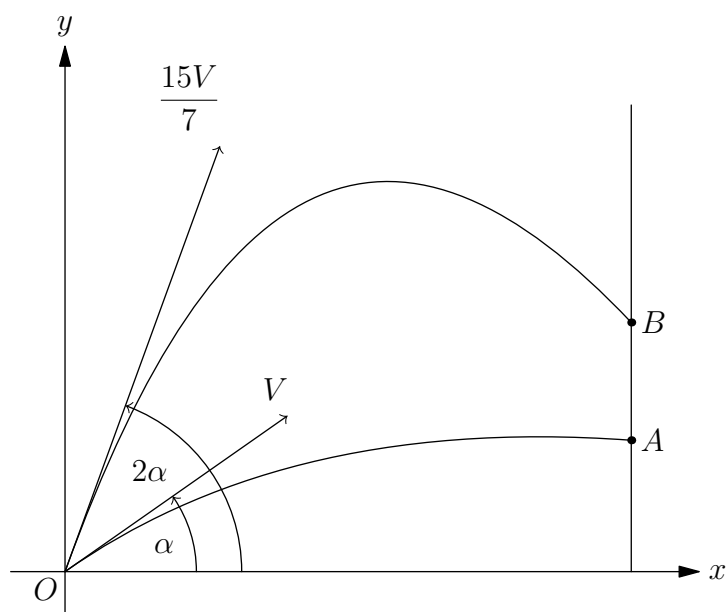
- (i) Show that $OMNL$ is a parallelogram. **2**
- (ii) Suppose that AX is perpendicular to MN . **4**
- Show that $|\underline{b}| = \sqrt{3}|\underline{a}|$.

Question 14 continues on page 12

Question 14 (continued)

- (d) Alex hits two balls at the same time from the origin O towards a target on a vertical wall. 3

The velocity of projection of the first ball is $V \text{ m s}^{-1}$ and its angle of projection is α to the horizontal, while the velocity of projection of the second ball is $\frac{15V}{7} \text{ m s}^{-1}$ and its angle of projection is 2α to the horizontal.



It is known that both balls hit the wall at the same time. By considering the position vectors for both balls, show that $\cos \alpha = \frac{5}{6}$.

- (e) *You may use the information in a standard normal distribution table to answer this question.* 4

W² education is holding a graduation party and sends out invitations to its students. All invited students are guaranteed to attend and each student is allowed to bring up to one *tag-along* (an additional student who was not invited).

The probability that any individual student will bring a *tag-along* is 40%.

Since the venue has a strict capacity limit of 400 total attendees due to space constraints, W² education deliberately invites less students in order to accommodate for any *tag-alongs*.

Determine the maximum number of students that W² education should invite in order to ensure that there is less than a 1% chance of the venue exceeding capacity.

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